

Vatsal Hitesh Lodaya

vatsallodaya04@gmail.com | +91 91674 10901 | Mulund, Mumbai
github.com/VATSALLODAYA27 | linkedin.com/in/vatsal-hitesh-lodaya | Portfolio: vatsal27.vercel.app

Profile

B.Tech Computer Engineer (Honours in Data Science) building full-stack web apps with the MERN stack and generative AI systems with LangGraph and RAG, with hands-on data analytics experience in SQL, Python and Power BI.

Education

Shah & Anchor Kutchhi Engineering College	2022-2026
Bachelor of Technology (B.Tech) Computer Engineering & Honours in Data Science	CGPA: 8

Experience

Associate Software Developer Intern Nimap Infotech LLP	June 2024 - July 2024
Assisted in web application development and debugging while collaborating with the development team on various software projects, gaining practical experience in the software and web development lifecycle.	

Projects

AI-Based Fish Detection System

- YOLO-based detection and classification of fish species from underwater images and live video, in a web app with sign-in, image upload and real-time detection. *(Python, Flask, YOLOv11, HTML, CSS, JavaScript, MySQL)*

Companio

- Find a verified person nearby for the same activity: PostGIS nearby search with privacy-safe map pins, OAuth, live Socket.IO chat; 254 unit and E2E tests, Docker. *(Next.js, NestJS, TypeScript, PostgreSQL, PostGIS, Prisma, Redis, Socket.IO, Docker)*

EDITH - Multi-Agent AI Orchestrator

- LangGraph supervisor that splits a request across specialist agents (RAG, docs, Excel, PPT, browser, email, calendar) in parallel, with human approval for risky actions. *(Python, LangGraph, LangChain, Groq, FastAPI, React, ChromaDB)*

RAG Assistant

- Chat with uploaded PDFs, with page-level citations; hybrid embedding + BM25 retrieval, reranking, OCR and resumable background indexing. *(React, Vite, FastAPI, Python, ChromaDB, Gemini, OpenRouter)*

Trade Advisor

- Rule-based market advisor with parallel LangGraph agents, scanner, no-lookahead backtester, option strike picker and PDF morning brief. *(Python, LangGraph, Streamlit, yfinance, Pandas, Plotly)*

Customer Churn Analysis Dashboard

- Analyzed 10,000+ customer records with SQL and Python to find churn drivers; Power BI dashboard cut manual reporting time by ~40%. *(SQL, Python, Pandas, Power BI)*

GYM Management System

- Manages memberships, training schedules, workout guidance and equipment purchases in one system. *(MERN Stack)*

Loan Default Prediction Model

- Compared Logistic Regression and Random Forest (87% accuracy); deployed as a Streamlit app for real-time predictions. *(Python, Scikit-learn, Streamlit)*

Credit Card Statement Parser

- Extracts bank, card holder, billing period, due date and total due from statement PDFs with per-bank regex. *(Python, Streamlit, pdfplumber, Regex)*

Skills

- Technical Skills:** C/C++, Python, SQL, MERN Stack, TypeScript, Next.js, FastAPI, Flask, Gen AI & LLMs, LangGraph, LangChain, RAG
- Other Software:** Power BI, Tableau, MS Tools, n8n, Docker, PostgreSQL, MySQL, Redis
- Strengths:** Rapid learning, public speaking, leadership under pressure

Certifications

- AWS Cloud Practitioner Essentials, AWS Training and Certification
- Paid Internship and Training Program in Web Development, Acmegrade
- Deloitte Australia Data Analytics Job Simulation on Forage, July 2025